Final Project - Bayesian Analysis of Alzheimers Data

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(Github: https://github.com/sarennalu/final-project-stat405.git)

Problem Formulation:

Alzheimer's disease is a progressive neurodegenerative disorder and the most common cause of dementia, characterized by declining memory and cognition. Early detection through biomarkers and MRI data is critical, as delayed treatment is associated with worse outcomes. In this project, we investigate clinical risk factors associated with Alzheimer's disease using Bayesian Statistics, aiming to identify which groups of people are at higher risk and support early intervention.

Our research questions are:

i) **How are brain volume, age and biological sex associated with the probability of receiving a diagnosis of Alzheimer's disease and the diagnosed CDR (Clinical dementia rating)**

- Is there an effect of sex on Alzheimer's disease probability?
- Do individuals with lower brain volume have higher CDR scores?
- Does higher education levels dampen the effect of brain volume on CDR?

Basic Biological Context and Important Variables

CDR (Clinical Dementia Rating) is a common tool used to assess cognitive impairment, with scores mapped as follows:
0: none   0.5: questionable   1: mild   2: moderate   3: severe
Any score > 0 indicates probable AD, making CDR both a diagnostic and severity measure.

MMSE: (Mini-Mental State Examination score) A brief test of global cognitive function, with the following scores describing the severity of cognitive decline.
24–30: Normal   19–23: Mild   10–18: Moderate   0–9: Severe

eTIV (estimated total intracranial volume) Estimated total volume inside skull, estimated from MRI results.

nWBV (normalized whole-brain volume) Proportion of the intracranial cavity occupied by brain tissue. A lower volume would indicate more brain atrophy. We are interested in this variable because brain atrophy is a key characteristic of Alzheimer's disease progression.

Educ (Education Level) describes an individual's highest level of education:
1: <HS   2: HS grad   3: some college   4: bachelor's   5: postgraduate

Key Modelling Challenges:

The key modelling challenge is that the response variable, CDR, is an ordinal variable, with higher values indicating higher disease severity. We will begin with naive models treating CDR as a binary random variable. Then we would build upon this with a more sophisticated model to include the ordering. This is grounded in the framework of progressively increasing model complexity, as often done in the course.

In addition, we will assess the posterior predictive probabilities for our model through cross-validation, a tool that we were introduced to in the course. We will apply concepts from the theme of Bayesian calibration in this section.

Literature Review

Neuroimaging studies show that brain volume decreases with age and more rapidly in individuals with AD, reflecting disease-related neurodegeneration. These structural changes are closely associated with clinical measures such as the Clinical Dementia Rating (CDR) and Mini-Mental State Examination (MMSE).

A widely used dataset for studying these relationships is the Open Access Series of Imaging Studies (OASIS) (Marcus et al., 2007), which combines MRI-derived brain measures with demographic and clinical variables. Key variables include age, sex, education, and imaging measures such as estimated total intracranial volume (eTIV) and normalized whole brain volume (nWBV). Previous studies using OASIS and related datasets have identified strong associations between brain atrophy and Alzheimer's disease severity (Fotenos et al., 2005; Buckner et al., 2004). In addition, machine learning approaches such as support vector machines (Klöppel et al., 2008) and deep learning models (Suk et al., 2014) have achieved high predictive accuracy using imaging features. However, these methods often prioritize prediction over interpretability and provide limited uncertainty quantification.

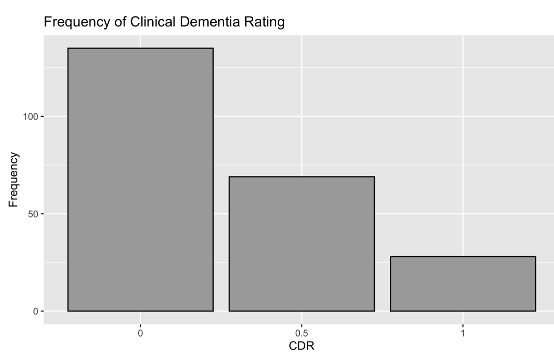
Bayesian statistical methods offer a complementary framework by providing full posterior distributions of model parameters, enabling direct uncertainty quantification and incorporation of prior knowledge. This is particularly valuable in medical settings, where uncertainty is critical for decision-making.

Bayesian approaches have been applied to Alzheimer's research to model disease progression and cognitive decline. For example, Fonteijn et al. (2012) proposed an event-based model describing disease progression as a sequence of latent events, while Donohue et al. (2014) used hierarchical models for longitudinal cognitive data. Despite these advantages, Bayesian methods remain relatively underutilized in analyses of the OASIS dataset, where most studies rely on frequentist or machine learning approaches.

In summary, by adopting a Bayesian perspective, this study contributes to the literature by providing interpretable, uncertainty-aware estimates of how brain structure and clinical variables jointly influence Alzheimer's disease risk and severity.

Exploratory Data Analysis

First, we must explore the dataset's feature distributions before designing and fitting models.

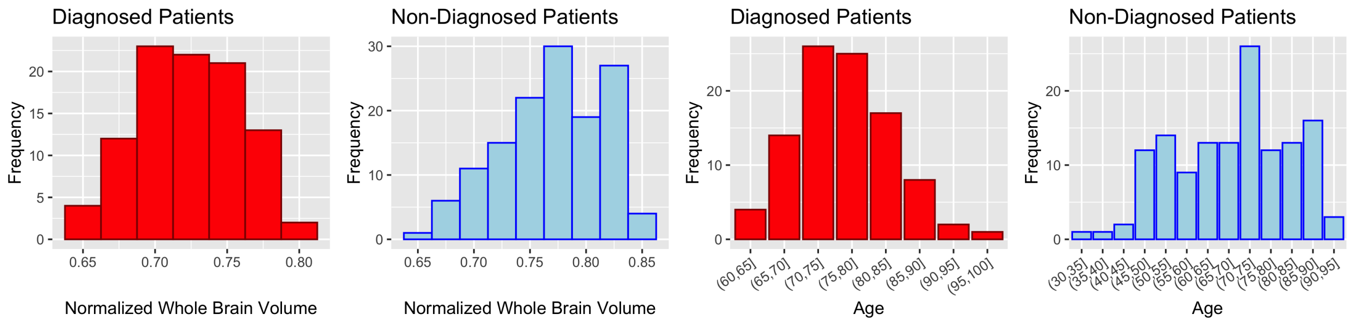


Most patients are not diagnosed with Alzheimers, and of the diagnosed patients, most of them have been diagnosed with only mild dementia.

Table 1: Diagnosis vs Sex

Biological_Sex	Diagnosed	NotDiagnosed
Female	57	97
Male	40	38

There are more females in the dataset, males have a higher diagnosis rate than females.



From the plot above, it appears diagnosed patients typically have less whole brain volume than non-diagnosed patients. Diagnosed patients tend to be older, where non diagnosed patients represent all ages.

Models

We build the following two models for our data. For the priors, standard deviations of the normal distributions have been kept small keeping in mind that covariates will be standardised.

Naive Bayesian Logistic Regression Model

$$\beta_j \sim \mathcal{N}(0, 3), \quad \text{for } j \in \{0, 1, 2, 3, 4, 43\}.$$

$$\begin{aligned} \eta_i = {} & \beta_0 + \beta_1 \cdot \text{sex}_{i,\text{male}} + \beta_2 \cdot \text{age}_i + \beta_3 \cdot \text{brain\_vol}_i + \beta_4 \cdot \text{educ}_i \\ & + \beta_{34} \cdot (\text{brain\_vol}_i \times \text{educ}_i). \end{aligned}$$

$$\text{CDR\_binary}_i \mid \eta_i \sim \text{Bernoulli}(logistic(\eta_i)).$$

Complex Model

Note that we have set $\beta_0 = 1$ for identifiability

$$\begin{aligned} \beta_j &\sim \mathcal{N}(0, 2) \quad (j = 1, 2, 3, 34) \\ \delta &\sim \text{Exp}\,(0.3) \\ k_0 &\sim \mathcal{N}(0, 1), \quad k_1 = k_0 + \delta \\ \sigma &\sim \text{Exponential}(0.3) \end{aligned}$$

Hidden Disease Severity Score

$$\begin{aligned} \mu(i) = {} & \beta_0 + \beta_1\,\text{sex\_male}_i + \beta_2\,\text{age}_i + \beta_3\,\text{brain\_vol}_i + \beta_4\,\text{educ}_i \\ & + \beta_{3,4}\,(\text{brain\_vol}_i \times \text{educ}_i) \end{aligned}$$

$$Z_i \mid \mu(i) \sim \mathcal{N}(\mu(i), \sigma)$$

Disease Outcome

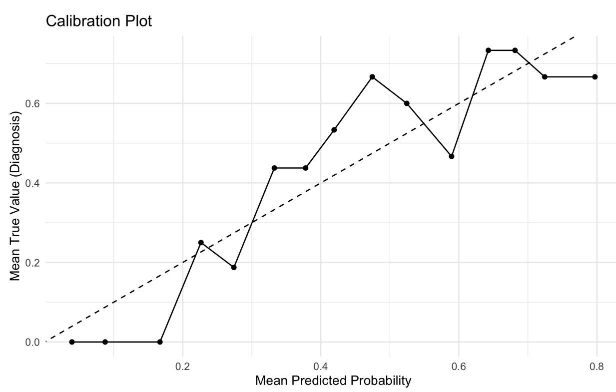
$$Y_i \mid Z_i = \begin{cases} 0 & \text{if } Z_i \leq k_0 \\ 0.5 & \text{if } k_0 \leq Z_i \leq k_1 \\ 1 & \text{if } Z_i > k_1 \end{cases}$$

Implementation of the Naive Model

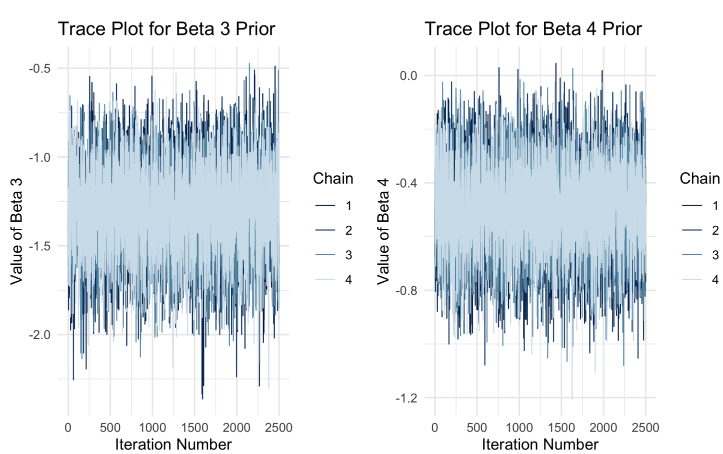
After fitting the model, we inspect the quality of the fit.

variable	mean	median	sd	mad	q5	q95	rhat	ess_bulk	ess_tail
lp__	-130.3020475	-129.9521250	1.7434656	1.5959077	-133.6149625	-128.115848	0.9999957	4014.000	5753.700
beta_0	1.3735945	1.3821760	1.9814129	1.9713963	-1.8674770	4.617477	1.0009348	4671.063	5136.355
beta_1	0.6363919	0.6345535	0.3282910	0.3308136	0.1028844	1.182588	1.0002459	6114.409	5174.584
beta_2	0.0144233	0.0140244	0.0163888	0.0162147	-0.0122494	0.042013	1.0002739	5109.413	5420.891
beta_3	-2.3942652	-2.3745006	2.2841749	2.2473440	-6.1081557	1.375524	1.0006261	4364.827	5052.990
beta_4	3.2911694	3.2811631	1.0223509	1.0184403	1.6269091	4.988054	1.0003153	3802.224	4736.381
beta_34	-4.9263032	-4.9299621	1.3879566	1.3829218	-7.2362108	-2.665211	1.0002284	3802.930	4881.806

`rhat` ≈ 1 for all parameters, indicating that the chains have all converged to the same distribution. The bulk and tail ESS are high, indicating efficient sampling.



As we can see, the model is only decently calibrated.



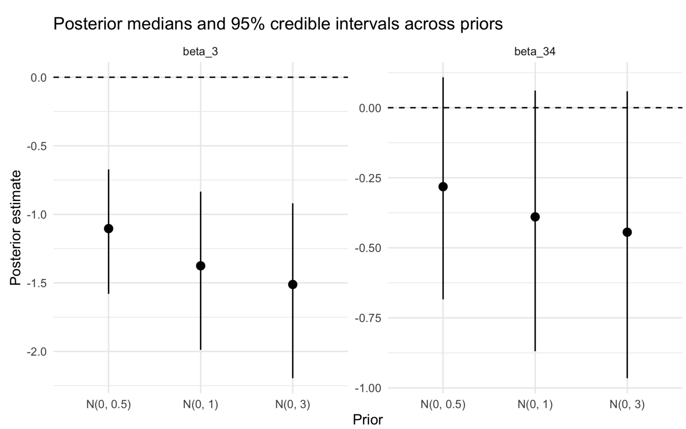
All 4 chains have plenty of overlap, and don't appear to be stuck at any point. This demonstrates good mixing for all of the priors shown. For the rank plots, see Appendix.

```
##
## Computed from 10000 by 232 log-likelihood matrix.
##
##          Estimate   SE
## elpd_loo   -129.1  7.6
## p_loo         6.1  0.7
## looic       258.2 15.1
## ------
## MCSE of elpd_loo is 0.0.
## MCSE and ESS estimates assume independent draws (r_eff=1).
##
## All Pareto k estimates are good (k < 0.7).
## See help('pareto-k-diagnostic') for details.
```

```
## Average probability assigned to true outcome: 0.573
```

The Pareto k estimates tell us that there are no outliers influencing the model.
The model is assigning 57.3% probability to the correct outcome (correct class of dementia diagnosis) on average for the leave-one-out predictions.

Sensitivity Analysis for Naive Model

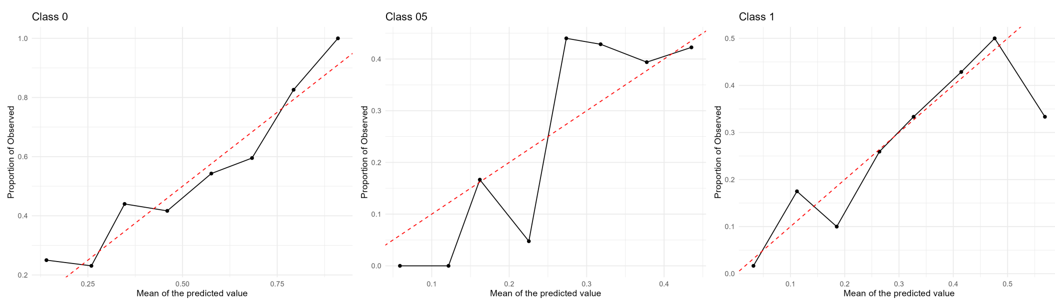


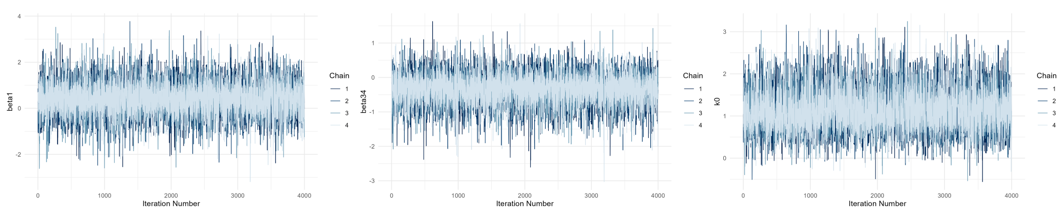
Implementation of the Complex Model

We now implement the complex model using STAN, once using its default algorithm and once using Variational Inference. After fitting using STAN's default algorithm (NUTS), we inspect the quality of the fit and calibration of the model.

variable	mean	median	sd	mad	q5	q95	rhat	ess_bulk	ess_tail
lp___	-182.6412980	-182.2613850	2.5612897	2.4478467	-187.309628	-179.1594335	1.001004	3345.428	6760.190
beta1	0.3948292	0.3863903	0.7147512	0.6538026	-0.776991	1.5766076	1.000827	10169.697	8113.184

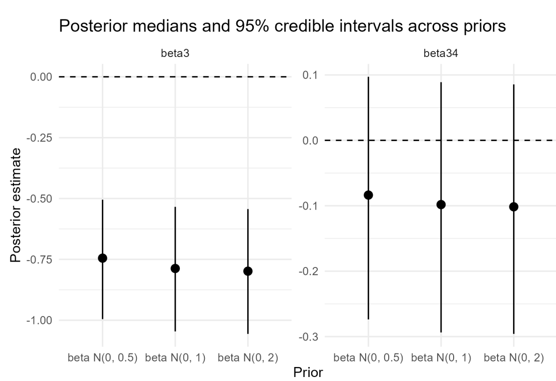
variable	mean	median	sd	mad	q5	q95	rhat	ess_bulk	ess_tail
beta2	-0.2910655	-0.2953637	0.5587883	0.5098471	-1.188606	0.6165874	1.000562	10369.118	9198.305
beta3	-3.1895708	-3.1107940	0.9943077	0.9861639	-4.958251	-1.6991747	1.001118	3361.062	3856.037
beta4	-1.1485452	-1.0823258	0.5125464	0.4818367	-2.082053	-0.4281073	1.000494	5472.295	8342.659
beta34	-0.3443083	-0.3235710	0.4307925	0.3909391	-1.083524	0.3260296	1.000417	10381.270	8908.580



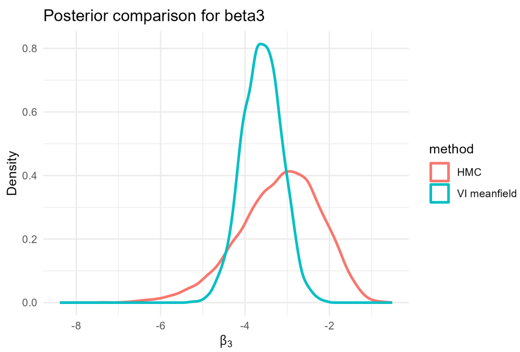


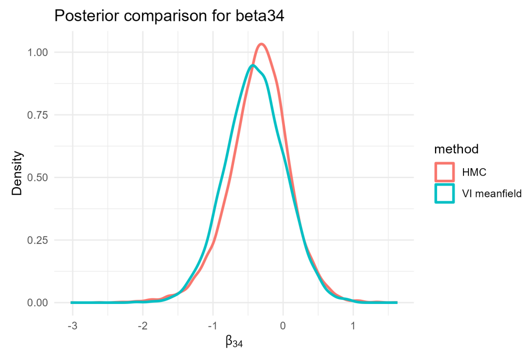
These plots demonstrate good mixing. For the rank plots, see Appendix. We also performed correctness of code checks (see Appendix).

Senstivity Analysis on the Complex Model



HMC vs VI on the Complex Model





Results and Discussion

We will now try to tackle some of the questions we were interested in using the complex model. We try to find if the coefficient of sex_male in our model greater than 0 using the following two ways:

1. *Posterior Probability*: We estimated $P(\beta_1 > 0|Data) = 0.726$. Additionally $\mathbb{E}(\beta_1|Data) = 0.395$ and a 90% credible interval is $[-0.777, 1.577]$. We have some evidence that keeping the other variables constant, biological males usually have a higher CDR score than females.

2. *Decision Theory Framework*: We wish to make a decision whether $\beta_1 > 0$. Let the action $a = 1$ mean concluding that there is a positive slope β_1. Define the loss function:

$$L(a, \beta_1) = 2\mathbb{I}(a = 1, \beta_1 < 0) + \mathbb{I}(a = 0, \beta > 0)$$

   where $a = 1$ indicates that we claim the slope is > 0. This loss function was chosen because we would like to penalise a false positive much more than a false negative. Using this loss function we estimate

$L(a) := \mathbb{E}(L(a, X)) = 0.548\mathbb{I}(a = 1) + 0.726\mathbb{I}(a = 0)$.

Hence, under this loss function, an optimal decision would be to conclude that the sex_male effect is positive.

Now, similarly to answer our second question we ask what is the probability of $\beta_3 + \beta_{34}educ_i < 0$? Using three different level of standardized education levels $(-1, 0, 1)$, we found that for each of the three levels, the posterior probability $Pr(\beta_3 + \beta_{34} * educ_i < 0|Data) \approx 1$. This table summarises the posterior means and credible intervals

Education Level	Posterior Mean of Brain Volume Effect	95% CrI	P(effect < 0)
Low (-1)	-2.845	(-5.015, -1.229)	1.00
Mean (0)	-3.190	(-5.355, -1.491)	1.00
High (-1)	-3.534	(-6.13, -1.565)	1.00

This gives strong evidence that that individuals with lower brain volume have higher CDR scores controlling for the other variables.

To answer our final question, we need to ask whether the interaction term in our complex model β_{34} is greater than 0. A positive term would mean that at higher education levels the effect of reducing brain volume on the severity of the disease is smaller.

1. *Posterior Probability*: We found $Pr(\beta_{34} > 0|Data) \approx 0.1948125$, which is very small and incdicates we are not very certain that this coefficient is positive.

2. *Decision Theory Framework*: Using the same loss function and actions as before, we calculated $L(a) := \mathbb{E}(L(a, X)) = 1.610375\mathbb{I}(a = 1) + 0.1948\mathbb{I}(a = 0)$, so the Bayes estimator would be to conclude that there isn't a dampening effect of education.

Prior Sensitivity Analysis

We evaluated robustness to prior variance. In the complex model, the posterior for β_3 remained stable across priors, with consistently negative credible intervals, indicating strong data influence. The interaction term β_{34} was also fairly stable but retained wider intervals, reflecting weaker evidence. In contrast, the naive model showed greater sensitivity: posterior estimates for β_3 and β_{34} shifted more and had wider intervals under different priors, indicating weaker identifiability and stronger prior dependence. Overall, the complex model yields more robust, data-driven inference.

Comparison of HMC and Variational Inference

We compared HMC with mean-field VI for the complex model. For β_3, VI produced a narrower posterior, underestimating uncertainty due to its independence assumptions, though both methods agreed on a negative effect. For β_{34}, the two approaches produced similar results, suggesting it is easier to approximate.

Given good HMC convergence diagnostics, its broader posterior is more reliable. VI captures posterior means well but underestimates uncertainty.

Limitations and Implication of Results

Our analysis has several limitations. We had limited data for CDR classes 0.5 and 1 and none above that, so we cannot conclude whether our model is appropriately specified across the full CDR range of 0–3. As undergraduates with limited medical knowledge, we assessed only the direction of main effects rather than clinically important thresholds, and our loss functions reflected our own assumptions rather than scientific evidence. We also included only one interaction term and assumed linear effects for continuous predictors, leaving potential misspecification unaddressed. Like other studies using the OASIS dataset, our findings are hypothesis-generating and require validation through longitudinal research.

Our results confirm that brain volume is positively associated with CDR scores, consistent with the broader biological literature. The finding that males face higher dementia risk (conditional on age, brain volume, and education) offers an interesting angle but should be interpreted cautiously, as it contradicts the general consensus that women are at greater risk. This discrepancy reflects the conditioned nature of our coefficients. Finally, we found no support for the protective effect of education on dementia severity.

Contributions of Each Group Member

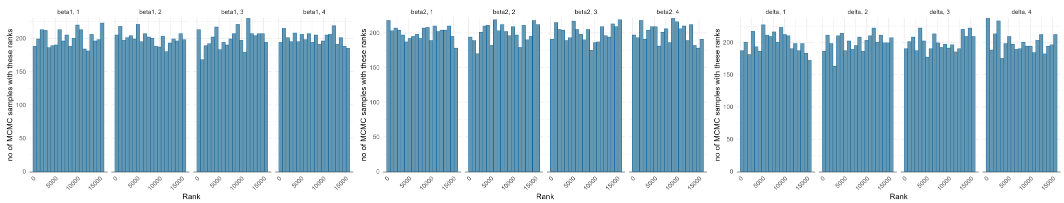
Each group member contributed roughly equally. Michael wrote the literature review, performed the sensitivity analysis for both models, and the variational inference for the complex model. Sachit wrote the problem formulation, primarily designed both models, and implemented the complex model, including the stan implementation, calibration checks, model diagnostics, evaluation of the posterior approximation, and the code correctness for both models. Sarenna set up the repository/virtual environment, processing the raw data and the exploratory data analysis, and implementing the naive model, including the stan implementation, calibration checks, model diagnostics, evaluation of the posterior approximation, and the leave-one-out for both models.

Appendix

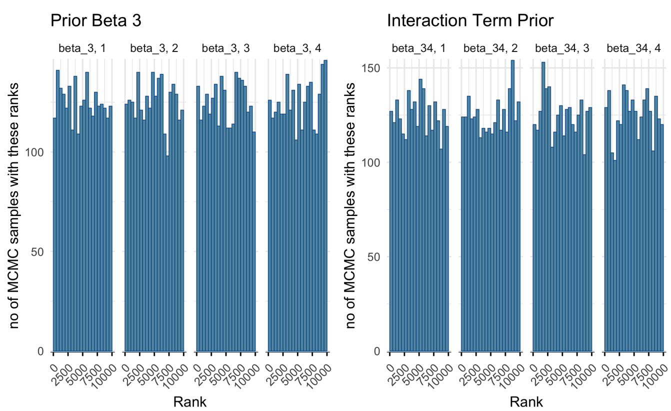
Sources:

1. https://www.alz.org/alzheimers-dementia/what-is-dementia
2. https://www.who.int/news-room/fact-sheets/detail/dementia

Rank plots for Complex Model



Rank plots for Naive Model



All of the histograms look roughly flat for all 4 chains for the parameters displayed. This is an indication of good mixing.

Correctness of Code Checks

We generate parameters and data according to our model many times. Then after fitting our model, what proportions of the times the 90%credible interval for the parameter is able to capture the true value of the parameter.

```
Niter = 30

for (i in 1:Niter) {

sd_beta = 2
```

```
beta1 = rnorm(1,mean =0,sd = sd_beta)
beta2 = rnorm(1, mean = 0, sd_beta)
beta3 = rnorm(1, mean = 0, sd_beta)
beta4 = rnorm(1, mean = 0, sd_beta)
beta34 = rnorm(1, mean = 0, sd_beta)
delta = rexp(1, 0.3)
sigma = rexp(1, 0.3)
k0 = rnorm(1, 0,1)

params = c(beta1, beta2, beta3, beta4,beta34, delta, sigma, k0)

get_disease = function(age, educ, nWBV, sex_male) {
  mu = beta1*sex_male +beta2*age + beta3*nWBV +beta4*educ +beta34*nWBV*educ
  Z = rnorm(1, mu, sd = sigma)
  if (Z <= k0) {
    return(0)
  } else if (Z <= k0 + delta) {
    return(0.5)
  } else {
    return(1)
  }
}

full_data <- predictors |>
  rowwise() |>
  mutate(y = get_disease(Age, Educ, nWBV, sex_male)) |>
  ungroup()

full_data

y = as.numeric(full_data$y)
age = as.numeric(full_data$Age)
sex = as.numeric(full_data$sex_male)
brain_volume =as.numeric(full_data$nWBV)
educ = as.numeric(full_data$Educ)
brain_educ = as.numeric(educ*brain_volume)

mod <- cmdstan_model("complex_model.stan")
fit = mod$sample(
  seed =405,
  chains = 1,
  refresh = 1000,
  data = list(
          y = y,
          N = nrow(full_data),
          sex = sex,
          age = age,
          brain_volume = brain_volume,
          educ = educ,
          brain_educ = brain_educ
        ),
  iter_sampling = 3000,
  show_messages = FALSE
```

```
)



true_values = c(beta1 = beta1,
                beta2 = beta2,
                beta3 = beta3,
                beta4 =beta4,
                beta34 = beta34,
                delta = delta,
                sigma = sigma,
                k0 =k0)

draws = fit$draws(variables = names(true_values))

summary_df = summarise_draws(
  draws,
  mean,
  sd,
  rhat,
  ess_bulk,
  ~quantile(., probs = c(0.025, 0.975))
)

summary_df = summary_df |> rename(lower= `2.5%`, upper = `97.5%`) |>
  mutate(true_value = true_values[variable],
         recovered= true_value <= upper & true_value >= lower, iteration = rep(i, 8))

simulation_df = bind_rows(simulation_df, summary_df)

}
```

```
simulation_results = simulation_df |>
  select(iteration, recovered, variable) |> mutate(recovered = as.numeric(recovered)) |> group_by(varial
simulation_results
```

variable	prob_recovered
beta1	0.9333333
beta2	1.0000000
beta3	0.9666667
beta34	0.9000000
beta4	0.9666667
delta	0.9333333
k0	0.9666667
sigma	0.9333333

We can see that for each parameter of the model proportion of the times each it is in the 90% credible interval is roughly 90%, so this confirms that our code can work as intended.

Stan File for Complex Model: HMC and VI

```
  data {
  int N;
  vector<lower=0, upper = 1>[N] y;
  vector<lower = 0, upper =1>[N] sex;
  vector[N] age;
  vector[N] brain_volume;
  vector[N] educ;
  vector[N] brain_educ;
}

parameters {
  real beta1;
  real beta2;
  real beta3;
  real beta4;
  real beta34;
  real k0;
  real<lower = 0> delta;
  real<lower = 0> sigma;
}

transformed parameters {
  real beta0 = 0;
  vector[N] mu = beta0 + beta1*sex + beta2*age
  + beta3*brain_volume + beta4*educ + beta34*brain_educ;
  real k1 = k0 + delta;
}

model {

  // prior
  beta1 ~ normal(0,2);
  beta2 ~ normal(0,2);
  beta3 ~ normal(0,2);
  beta4 ~ normal(0,2);
  beta34 ~ normal(0,2);
  delta ~ exponential(0.3);
  sigma ~ exponential(0.3);
  k0 ~ normal(0,1);

  for (i in 1:N) {
    if (y[i] == 0) {
      target += normal_lcdf(k0| mu[i], sigma);
    } else if (y[i] == 0.5) {
    target += log_diff_exp(
                normal_lcdf(k1 | mu[i], sigma),
                normal_lcdf(k0 | mu[i], sigma));
    } else {
      target += normal_lccdf(k1| mu[i], sigma);
    }
  }
```

```
}

generated quantities {
  vector[N] p0;
  vector[N] p05;
  vector[N] p1;
  vector[N] log_lik;

  for (i in 1:N) {
    p0[i] = normal_cdf(k0| mu[i], sigma);
    p05[i] = normal_cdf(k1| mu[i], sigma) - normal_cdf(k0| mu[i], sigma);
    p1[i] =  1- normal_cdf(k1|mu[i], sigma);

    if (y[i] == 0) {
      log_lik[i] = normal_lcdf(k0 | mu[i], sigma);
    } else if (y[i] == 0.5) {
      log_lik[i] = log_diff_exp(
        normal_lcdf(k1 | mu[i], sigma),
        normal_lcdf(k0 | mu[i], sigma)
      );
    } else {
      log_lik[i] = normal_lccdf(k1 | mu[i], sigma);
    }
  }
}
```

HMC vs VI implementation

```
data_processed <- read.csv("../../data/processed/oasis_processed.csv")

data_processed <- data_processed |>
  select(Age, Educ, nWBV, CDR, sex_male) |>
  filter(
    !is.na(Age),
    !is.na(Educ),
    !is.na(nWBV),
    !is.na(CDR),
    !is.na(sex_male)
  )

N <- nrow(data_processed)
y <- data_processed$CDR
age <- data_processed$Age
sex <- data_processed$sex_male
brain_volume <- data_processed$nWBV
educ <- data_processed$Educ

# Standardise continuous predictors
age_z <- as.numeric(scale(age))
brain_volume_z <- as.numeric(scale(brain_volume))
educ_z <- as.numeric(scale(educ))
brain_educ_z <- brain_volume_z * educ_z
```

```r
stan_data <- list(
  N = N,
  y = y,
  sex = sex,
  age = age_z,
  brain_volume = brain_volume_z,
  educ = educ_z,
  brain_educ = brain_educ_z
)

mod <- cmdstan_model("complex_model.stan")
fit_hmc = mod$sample(
  seed =405,
  chains = 4,
  refresh = 1000,
  data = stan_data,
  iter_sampling = 4000,
  show_messages = FALSE
)

fit_hmc$summary(c("beta1", "beta2", "beta3", "beta4", "beta34", "k0", "delta", "sigma"))

fit_vi <- mod$variational(
  data = stan_data,
  seed = 405,
  algorithm = "meanfield",
  iter = 20000,
  grad_samples = 1,
  elbo_samples = 100,
  output_samples = 4000,
)

fit_vi$summary(c("beta1", "beta2", "beta3", "beta4", "beta34", "k0", "delta", "sigma"))

params <- c("beta1", "beta2", "beta3", "beta4", "beta34", "k0", "delta", "sigma")

draws_hmc <- as_draws_df(fit_hmc$draws(params))
draws_vi  <- as_draws_df(fit_vi$draws(params))

make_summary <- function(draws_df, method_name, params) {
  data.frame(
    parameter = params,
    mean = sapply(params, function(p) mean(draws_df[[p]])),
    median = sapply(params, function(p) median(draws_df[[p]])),
    sd = sapply(params, function(p) sd(draws_df[[p]])),
    q2.5 = sapply(params, function(p) quantile(draws_df[[p]], 0.025)),
    q97.5 = sapply(params, function(p) quantile(draws_df[[p]], 0.975)),
    prob_negative = sapply(params, function(p) mean(draws_df[[p]] < 0)),
    method = method_name
  )
}

summary_comparison <- rbind(
```

```r
  make_summary(draws_hmc, "HMC", params),
  make_summary(draws_vi, "VI meanfield", params)
)

summary_comparison

plot_df_beta3 <- rbind(
  data.frame(value = draws_hmc$beta3, method = "HMC"),
  data.frame(value = draws_vi$beta3, method = "VI meanfield")
)

p_beta3 <- ggplot(plot_df_beta3, aes(x = value, colour = method)) +
  geom_density(linewidth = 1) +
  labs(
    title = "Posterior comparison for beta3",
    x = expression(beta[3]),
    y = "Density"
  )

p_beta3

ggsave("../../plots/complex_model/beta3_density_hmc_vi.png", plot = p_beta3, width = 6, height = 4, dpi

plot_df_beta34 <- rbind(
  data.frame(value = draws_hmc$beta34, method = "HMC"),
  data.frame(value = draws_vi$beta34, method = "VI meanfield")
)

p_beta34 <- ggplot(plot_df_beta34, aes(x = value, colour = method)) +
  geom_density(linewidth = 1) +
  labs(
    title = "Posterior comparison for beta34",
    x = expression(beta[34]),
    y = "Density"
  )

p_beta34

ggsave("../../plots/complex_model/beta34_density_hmc_vi.png", plot = p_beta34, width = 6, height = 4, d
```

Calibration and Diagnostics of Complex Model

```r
data_processed = read.csv("../../data/processed/oasis_processed.csv")

data_processed = data_processed |> select(Age, Educ, nWBV, CDR, sex_male)

head(data_processed)

N = nrow(data_processed)
y = data_processed$CDR
age = data_processed$Age
```

```r
sex = data_processed$sex_male
brain_volume =data_processed$nWBV
educ = data_processed$Educ
brain_educ = data_processed$Educ*data_processed$nWBV

## Standardising
age_z = as.numeric(scale(age))
brain_volume_z = as.numeric(scale(brain_volume))
educ_z = as.numeric(scale(educ))
brain_educ_z = brain_volume_z*educ_z

mod <- cmdstan_model("complex_model.stan")

fit = mod$sample(
  seed =405,
  chains = 4,
  refresh = 1000,
  data = list(
          y = y,
          N = nrow(data_processed),
          sex = sex,
          age = age_z,
          brain_volume = brain_volume_z,
          educ = educ_z,
          brain_educ = brain_educ_z
        ),
  iter_sampling = 4000,
  show_messages = FALSE
)

fit
fit_summary <- fit$summary()
write.csv(fit_summary, "../../results/complex_model/fit_summary.csv", row.names = FALSE)


summary_df <- fit$summary()
write.csv(summary_df, "fit_summary_complex_model.csv", row.names = FALSE)

theme_set(theme_minimal())

p_beta1 <- mcmc_trace(fit$draws("beta1"))  + xlab("Iteration Number")
p_k0 <- mcmc_trace(fit$draws("k0"))     + xlab("Iteration Number")
p_delta <- mcmc_trace(fit$draws("delta"))  + xlab("Iteration Number")
p_beta34 <- mcmc_trace(fit$draws("beta34")) + xlab("Iteration Number")

p_beta1
p_k0
p_delta
p_beta34

ggsave("../../plots/plots_complex_model/beta1_trace.png",
       plot = p_beta1, width = 7, height = 4, dpi = 300)
ggsave("../../plots/plots_complex_model/k0_trace.png",
```

```r
       plot = p_k0, width = 7, height = 4, dpi = 300)
ggsave("../../plots/plots_complex_model/delta_trace.png",
       plot = p_delta, width = 7, height = 4, dpi = 300)
ggsave("../../plots/plots_complex_model/beta34_trace.png",
       plot = p_beta34, width = 7, height = 4, dpi = 300)


rank_hist_beta1 <- mcmc_rank_hist(fit$draws("beta1")) +
                      ylab("no of MCMC samples with these ranks") +
                      theme_minimal() +
                      theme(axis.text.x = element_text(angle = 45, hjust = 1))

rank_hist_beta2 <- mcmc_rank_hist(fit$draws("beta2")) +
                      ylab("no of MCMC samples with these ranks") +
                      theme_minimal() +
                      theme(axis.text.x = element_text(angle = 45, hjust = 1))

rank_hist_beta34 <- mcmc_rank_hist(fit$draws("beta34")) +
                      ylab("no of MCMC samples with these ranks") +
                      theme_minimal() +
                      theme(axis.text.x = element_text(angle = 45, hjust = 1))

rank_hist_k0 <- mcmc_rank_hist(fit$draws("k0")) +
                  ylab("no of MCMC samples with these ranks") +
                  theme_minimal() +
                  theme(axis.text.x = element_text(angle = 45, hjust = 1))

rank_hist_delta <- mcmc_rank_hist(fit$draws("delta")) +
                      ylab("no of MCMC samples with these ranks") +
                      theme_minimal() +
                      theme(axis.text.x = element_text(angle = 45, hjust = 1))

rank_hist_beta1
rank_hist_beta2
rank_hist_beta34
rank_hist_k0
rank_hist_delta

ggsave("../../plots/plots_complex_model/beta1_rank.png",
       plot = rank_hist_beta1, width = 7, height = 4, dpi = 300)
ggsave("../../plots/plots_complex_model/beta2_rank.png",
       plot = rank_hist_beta2, width = 7, height = 4, dpi = 300)
ggsave("../../plots/plots_complex_model/beta34_rank.png",
       plot = rank_hist_beta34, width = 7, height = 4, dpi = 300)
ggsave("../../plots/plots_complex_model/k0_rank.png",
       plot = rank_hist_k0, width = 7, height = 4, dpi = 300)
ggsave("../../plots/plots_complex_model/k0_rank.png",
       plot = rank_hist_delta, width = 7, height = 4, dpi = 300)


pred_0 = colMeans(as_draws_matrix(fit$draws("p0")))
pred_05= colMeans(as_draws_matrix(fit$draws("p05")))
pred_1 =  colMeans(as_draws_matrix(fit$draws("p1")))
```

```
calibration_check = function(pred_prob, observed_outcome, bins) {
  df = data.frame(predicted = pred_prob,
                  observed = observed_outcome)

  df$bin = cut(df$predicted, breaks = bins)

  summary = df |> group_by(bin) |>
           summarise(mean_predicted = mean(predicted),
                     mean_observed = mean(observed),
                     n = n())

  return(summary)

}


observed_0 = as.numeric(data_processed$CDR == 0)
observed_1 = as.numeric(data_processed$CDR == 1)
observed_05= as.numeric(data_processed$CDR == 0.5)

c0 = calibration_check(pred_0, observed_0, bins = 8)
c1 = calibration_check(pred_1, observed_1, bins = 8)
c05 = calibration_check(pred_05, observed_05, bins = 8)

c0
c1
c05

calibration_plot = function(cal_df, title) {
  ggplot(cal_df, aes(x = mean_predicted, y = mean_observed)) +
  geom_point() +
  geom_line() +
  labs(x = "Mean of the predicted value", y = "Proportion of Observed") +
  geom_abline(intercept = 0, slope = 1,
              linetype = "dashed", color = "red") +
  ggtitle(title)
}


p_c0  <- calibration_plot(c0, "Class 0")
p_c1  <- calibration_plot(c1, "Class 1")
p_c05 <- calibration_plot(c05, "Class 05")

p_c0
p_c1
p_c05

ggsave("../../plots/plots_complex_model/calibration_class0.png",
       plot = p_c0, width = 6, height = 5, dpi = 300)
ggsave("../../plots/plots_complex_model/calibration_class1.png",
       plot = p_c1, width = 6, height = 5, dpi = 300)
ggsave("../../plots/plots_complex_model/calibration_class05.png",
```

```
      plot = p_c05, width = 6, height = 5, dpi = 300)
```

Sensitivity Analysis Naive Model

```
data {
  int<lower=1> N;
  vector[N] sex_male;
  vector[N] age;
  vector[N] brain_vol;
  vector[N] educ;
  array[N] int<lower=0, upper=1> CDR_binary;

  real<lower=0> prior_sd;       // slope prior sd
  real<lower=0> intercept_sd;   // intercept prior sd
}

transformed data {
  vector[N] brain_educ = brain_vol .* educ;
}

parameters {
  real beta_0;
  real beta_1;
  real beta_2;
  real beta_3;
  real beta_4;
  real beta_34;
}

transformed parameters {
  vector[N] eta;
  vector[N] theta;

  eta = beta_0
      + beta_1 * sex_male
      + beta_2 * age
      + beta_3 * brain_vol
      + beta_4 * educ
      + beta_34 * brain_educ;

  theta = inv_logit(eta);
}

model {
  // Priors
  beta_0  ~ normal(0, intercept_sd);
  beta_1  ~ normal(0, prior_sd);
  beta_2  ~ normal(0, prior_sd);
  beta_3  ~ normal(0, prior_sd);
  beta_4  ~ normal(0, prior_sd);
  beta_34 ~ normal(0, prior_sd);
```

```
  // Likelihood
  CDR_binary ~ bernoulli(theta);
}

generated quantities {
  vector[N] log_lik;
  vector[N] p_pred;
  array[N] int y_pred;

  for (i in 1:N) {
    p_pred[i] = theta[i];
    y_pred[i] = bernoulli_rng(p_pred[i]);
    log_lik[i] = bernoulli_lpmf(CDR_binary[i] | theta[i]);
  }
}
```

```
data_processed <- read.csv("../../data/processed/oasis_processed.csv")

data <- data_processed %>%
  select(sex_male, Age, nWBV, Educ, CDR_binary) %>%
  filter(
    Educ > 1,
    Educ < 5,
    Age > 0,
    nWBV > 0,
    sex_male %in% c(0, 1),
    CDR_binary %in% c(0, 1)
  ) %>%
  mutate(
    age_z = as.numeric(scale(Age)),
    brain_vol_z = as.numeric(scale(nWBV)),
    educ_z = as.numeric(scale(Educ))
  )

stan_data_base <- list(
  N = nrow(data),
  sex_male = data$sex_male,
  age = data$age_z,
  brain_vol = data$brain_vol_z,
  educ = data$educ_z,
  CDR_binary = data$CDR_binary
)

naive_model <- cmdstan_model("naive_model_sensitivity.stan")

prior_grid <- tibble(
  prior = c("N(0, 0.5)", "N(0, 1)", "N(0, 3)"),
  prior_sd = c(0.5, 1.0, 3.0),
  intercept_sd = c(2.0, 2.0, 3.0)
)

fit_one_model <- function(prior_sd, intercept_sd, prior) {
  fit <- naive_model$sample(
```

```r
    seed = 405,
    chains = 4,
    parallel_chains = 4,
    iter_sampling = 2500,
    show_messages = FALSE,
    data = c(
      stan_data_base,
      list(prior_sd = prior_sd, intercept_sd = intercept_sd)
    )
  )

  list(prior = prior, fit = fit)
}

fit_list <- pmap(prior_grid, fit_one_model)
names(fit_list) <- prior_grid$prior

params <- c("beta_0", "beta_1", "beta_2", "beta_3", "beta_4", "beta_34")

extract_summary <- function(fit_obj, prior_name) {
  draws <- as_draws_df(fit_obj$draws(params))

  tibble(
    parameter = params,
    mean = sapply(params, function(p) mean(draws[[p]])),
    median = sapply(params, function(p) median(draws[[p]])),
    sd = sapply(params, function(p) sd(draws[[p]])),
    q2.5 = sapply(params, function(p) quantile(draws[[p]], 0.025)),
    q97.5 = sapply(params, function(p) quantile(draws[[p]], 0.975)),
    prob_negative = sapply(params, function(p) mean(draws[[p]] < 0)),
    prior = prior_name
  )
}

summary_table <- map2_dfr(
  fit_list,
  names(fit_list),
  ~ extract_summary(.x$fit, .y)
)

summary_table %>% arrange(parameter, prior)

diagnostic_table <- map2_dfr(
  fit_list,
  names(fit_list),
  function(x, nm) {
    x$fit$summary(params) %>%
      as_tibble() %>%
      mutate(prior = nm) %>%
      select(prior, variable, mean, sd, q5, q95, rhat, ess_bulk, ess_tail)
  }
)
```

```r
diagnostic_table

posterior_beta3 <- map2_dfr(
  fit_list,
  names(fit_list),
  function(x, nm) {
    draws <- as_draws_df(x$fit$draws("beta_3"))
    tibble(value = draws$beta_3, prior = nm)
  }
)

ggplot(posterior_beta3, aes(x = value, color = prior)) +
  geom_density(linewidth = 1) +
  labs(
    title = "Posterior density of brain-volume effect across priors",
    x = expression(beta[3]),
    y = "Density",
    color = "Prior"
  )
ggsave("../../plots/naive_model/posterior_beta3.png")

posterior_beta34 <- map2_dfr(
  fit_list,
  names(fit_list),
  function(x, nm) {
    draws <- as_draws_df(x$fit$draws("beta_34"))
    tibble(value = draws$beta_34, prior = nm)
  }
)

ggplot(posterior_beta34, aes(x = value, color = prior)) +
  geom_density(linewidth = 1) +
  labs(
    title = "Posterior density of interaction effect across priors",
    x = expression(beta[34]),
    y = "Density",
    color = "Prior"
  )
ggsave("../../plots/naive_model/posterior_beta34.png")

interval_plot_data <- summary_table %>%
  filter(parameter %in% c("beta_3", "beta_34"))

ggplot(interval_plot_data,
       aes(x = prior, y = median, ymin = q2.5, ymax = q97.5)) +
  geom_pointrange() +
  facet_wrap(~ parameter, scales = "free_y") +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(
    title = "Posterior medians and 95% credible intervals across priors",
    x = "Prior",
    y = "Posterior estimate"
  )
```

```r
ggsave("../../plots/naive_model/posterior_medians.png")
```

Sensitivity Analysis Complex Model

```stan
data {
  int<lower=1> N;
  array[N] int<lower=1, upper=3> y_cat;

  vector[N] sex;
  vector[N] age;
  vector[N] brain_volume;
  vector[N] educ;
  vector[N] brain_educ;

  real<lower=0> beta_sd;
  real<lower=0> c1_sd;
  real<lower=0> delta_rate;
}

parameters {
  real beta1;
  real beta2;
  real beta3;
  real beta4;
  real beta34;

  real c1;
  real<lower=0> delta;
}

transformed parameters {
  vector[N] eta;
  real c2;

  eta = beta1 * sex
      + beta2 * age
      + beta3 * brain_volume
      + beta4 * educ
      + beta34 * brain_educ;

  c2 = c1 + delta;
}

model {
  beta1  ~ normal(0, beta_sd);
  beta2  ~ normal(0, beta_sd);
  beta3  ~ normal(0, beta_sd);
  beta4  ~ normal(0, beta_sd);
  beta34 ~ normal(0, beta_sd);

  c1    ~ normal(0, c1_sd);
```

```
  delta ~ exponential(delta_rate);

  for (i in 1:N) {
    if (y_cat[i] == 1) {
      target += normal_lcdf(c1 | eta[i], 1);
    } else if (y_cat[i] == 2) {
      target += log_diff_exp(
        normal_lcdf(c2 | eta[i], 1),
        normal_lcdf(c1 | eta[i], 1)
      );
    } else {
      target += normal_lccdf(c2 | eta[i], 1);
    }
  }
}

generated quantities {
  vector[N] p1;
  vector[N] p2;
  vector[N] p3;
  vector[N] log_lik;

  for (i in 1:N) {
    p1[i] = normal_cdf(c1 | eta[i], 1);
    p2[i] = normal_cdf(c2 | eta[i], 1) - normal_cdf(c1 | eta[i], 1);
    p3[i] = 1 - normal_cdf(c2 | eta[i], 1);

    if (y_cat[i] == 1) {
      log_lik[i] = normal_lcdf(c1 | eta[i], 1);
    } else if (y_cat[i] == 2) {
      log_lik[i] = log_diff_exp(
        normal_lcdf(c2 | eta[i], 1),
        normal_lcdf(c1 | eta[i], 1)
      );
    } else {
      log_lik[i] = normal_lccdf(c2 | eta[i], 1);
    }
  }
}
```

```
data <- data_processed %>%
  select(Age, Educ, nWBV, CDR, sex_male) %>%
  filter(
    !is.na(Age),
    !is.na(Educ),
    !is.na(nWBV),
    !is.na(CDR),
    !is.na(sex_male),
    Age > 0,
    nWBV > 0,
    sex_male %in% c(0, 1),
    CDR %in% c(0, 0.5, 1)
  ) %>%
```

```r
  mutate(
    age_z = as.numeric(scale(Age)),
    brain_volume_z = as.numeric(scale(nWBV)),
    educ_z = as.numeric(scale(Educ)),
    brain_educ_z = brain_volume_z * educ_z,
    y_cat = case_when(
      CDR == 0   ~ 1L,
      CDR == 0.5 ~ 2L,
      CDR == 1   ~ 3L
    )
  )

stan_data_base <- list(
  N = nrow(data),
  y_cat = data$y_cat,
  sex = as.numeric(data$sex_male),
  age = data$age_z,
  brain_volume = data$brain_volume_z,
  educ = data$educ_z,
  brain_educ = data$brain_educ_z
)

mod <- cmdstan_model("complex_model_sensitivity.stan")

prior_grid <- tibble(
  prior = c("beta N(0, 0.5)", "beta N(0, 1)", "beta N(0, 2)"),
  beta_sd = c(0.5, 1.0, 2.0),
  c1_sd = c(1.0, 1.0, 1.0),
  delta_rate = c(0.3, 0.3, 0.3)
)

fit_one_complex <- function(prior, beta_sd, c1_sd, delta_rate) {
  fit <- mod$sample(
    seed = 405,
    chains = 4,
    iter_sampling = 4000,
    refresh = 1000,
    show_messages = FALSE,
    data = c(
      stan_data_base,
      list(
        beta_sd = beta_sd,
        c1_sd = c1_sd,
        delta_rate = delta_rate
      )
    )
  )

  list(prior = prior, fit = fit)
}

fit_list <- pmap(prior_grid, fit_one_complex)
names(fit_list) <- prior_grid$prior
```

```r
diagnostic_table <- map2_dfr(
  fit_list,
  names(fit_list),
  function(x, nm) {
    x$fit$summary(params) %>%
      as_tibble() %>%
      mutate(prior = nm) %>%
      select(prior, variable, mean, sd, q5, q95, rhat, ess_bulk, ess_tail)
  }
)

diagnostic_table

posterior_beta3 <- map2_dfr(
  fit_list,
  names(fit_list),
  function(x, nm) {
    draws <- as_draws_df(x$fit$draws("beta3"))
    tibble(value = draws$beta3, prior = nm)
  }
)

posterior_beta3 <- ggplot(posterior_beta3, aes(x = value, color = prior)) +
  geom_density(linewidth = 1) +
  labs(
    title = "Posterior density of brain-volume effect across priors",
    x = expression(beta[3]),
    y = "Density",
    color = "Prior"
  )

posterior_beta3

ggsave("../../plots/plots_complex_model/posterior_beta3_sensitivity.png",
       plot = posterior_beta3, width = 6, height = 4, dpi = 300)

posterior_beta34 <- map2_dfr(
  fit_list,
  names(fit_list),
  function(x, nm) {
    draws <- as_draws_df(x$fit$draws("beta34"))
    tibble(value = draws$beta34, prior = nm)
  }
)

posterior_beta34 <- ggplot(posterior_beta34, aes(x = value, color = prior)) +
  geom_density(linewidth = 1) +
  labs(
    title = "Posterior density of interaction effect across priors",
    x = "beta34",
    y = "Density",
    color = "Prior"
  )
```

```r
posterior_beta34

ggsave("../../plots/plots_complex_model/posterior_beta34_sensitivity.png",
       plot = posterior_beta3, width = 6, height = 4, dpi = 300)

interval_plot_data <- summary_table %>%
  filter(parameter %in% c("beta3", "beta34", "c1", "delta", "c2"))

cred_interval <- ggplot(
  interval_plot_data,
  aes(x = prior, y = median, ymin = q2.5, ymax = q97.5)
) +
  geom_pointrange() +
  facet_wrap(~ parameter, scales = "free_y") +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(
    title = "Posterior medians and 95% credible intervals across priors",
    x = "Prior",
    y = "Posterior estimate"
  )

cred_interval

ggsave("../../plots/plots_complex_model/cred_interval_sensitivity.png",
       plot = cred_interval, width = 6, height = 4, dpi = 300)
```